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The involvement of experts in Chinese policymaking: co-production of China's polar science and policy

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ABSTRACT

Using polar policy as a case, this article examines experts' involvement in Chinese foreign policymaking. Building on research on the party-state's internal consultative information system, it shows how Chinese polar experts are involved in polar policymaking through state institutions and party information channels. The article draws on 19 semi-structured interviews with Chinese polar experts and policy professionals conducted in China in April 2025, which were supplemented by institutional documents and publicly available sources. The findings distinguish three practices: horizontal advisory bodies mandated to coordinate expertise across bureaucratic systems; ad hoc mobilisation of expertise for major policy inputs, such as in the drafting of the 2018 Arctic Policy White Paper; and routine feedback and issue-framing through internal reference reporting (*neican*) and leaders' written instructions (*pishi*), which increasingly reward report-writing as a form of political and organisational capital. The study extends Sinological scholarship on the internal consultative information system to polar policy and specifies how expert knowledge reaches decision-makers. The study also theorises the *neican–pishi* system as boundary infrastructure between science and policy and as a site of experts' boundary spanning. The expansion of party reporting channels as boundary infrastructure broadens participation and increases competition and screening. This shows how expertise is produced, performed, and governed in contemporary China.

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China's Arctic policy; internal reference reporting (neican); leaders' instructions (pishi); expertise; co-production; boundary infrastructure

Introduction

Since the early 2010s, the Communist Party of China (CCP) has consolidated a vertically integrated 'internal consultative information system' centred on the General Offices of the CCP, which receive, filter, and transmit internal reference reports (*neican, zhuanbao*) upward and implement leaders' instructions (*pishi*) downward.[1] IR scholars and polar experts are also increasingly involved in the creation of exclusive analyses on different political issues. Polar experts in China work at universities, research institutes, and think

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[1]Tsai and Liao, "Concentrating Power to Accomplish Big Things," 297–310; Tsai and Liao, "The Authority, Functions, and Political Intrigues of the General Office of the Chinese Communist Party," 46–67; and e and Xue, "Motivated to Be Policy Entrepreneurs," 1–13.

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tanks and report through ministries that they are administratively subordinate to, as well as through party organs that bypass ministerial hierarchies. This system thus runs in parallel with the traditional vertical functional systems (*xitong*), which are well-described in the literature.[2]

Scholarship describes China's polar policymaking as coherent and coordinated, despite its fragmented character.[3] It is argued that 'China follows a comprehensive approach designed to advance the party-state's influence in the Arctic' across political, economic, scientific, and military arms.[4] China's polar science serves national strategy, and polar expeditions, icebreakers, observation networks, and remote sensing support scientific, commercial, governance, defence, and status objectives.[5] Chinese scholarly publications have become a valuable source of knowledge about policy thinking in Beijing.[6] Social science is often treated as 'part of the CCP propaganda system', with a difference between internal and external messaging (*neiwai youbie*).[7] Experts may sometimes have influence 'bottom up'.[8] This literature, however, has largely accepted the party-state's own description of the process as orchestrated. The mechanisms by which expert knowledge is converted into policy remain understudied.

This article explores the co-production of China's polar policy consultation regime, especially the way polar expertise is commissioned, produced, reviewed, and rewarded. In China, decision-making authority is centralised, and research aligns with the party-state's priorities. However, because polar governance issues are complex, interdisciplinary, and international, the party-state centre cannot fully specify its knowledge needs in advance. Experts supply some of the terms used to pose the problem and use available consultation mechanisms to anticipate the party-state's knowledge needs. The article also traces how this regime has changed over time.

This article seeks to make three contributions to the literature. First, it extends recent work on China's internal consultative information system to foreign policy, using polar policy as a case. As such, it adds to the literature on how experts in 'new-type think tanks' act as hidden policy entrepreneurs in foreign policy. Second, it specifies channels of expert knowledge involvement and shows how these channels create different incentives for expert involvement. Third, it advances understanding of polar expertise in China and uses this case to theorise the *neican–pishi* system as a boundary infrastructure and experts' boundary spanning between policy and expertise.

The article is based on 19 semi-structured interviews with Chinese polar experts conducted in China in April 2025. Interviewees include social and natural scientists, university staff, think-tank and private sector consultants, and *neican* report contributors or editors.[9] All interviews were conducted confidentially, and interviews are anonymised. The first-hand perspectives of Chinese polar experts provide a strong basis for searching documents and other publicly available sources that guide and supplement professionals'

[2]Brady, *China as a Polar Great Power*, 113.
[3]Brady, *China as a Polar Great Power*; and Kossa, "China and Arctic Science in the Xi Jinping Era."
[4]Puranen and Kopra, "China's Arctic Strategy."
[5]Brady, *China as a Polar Great Power*; Puranen and Kopra, "China's Arctic Strategy"; and Kossa, "China and Arctic Science in the Xi Jinping Era."
[6]Mokry, *Chinese Scholars and Think Tanks' Constructions of China's National Interest.*
[7]Brady, *China as a Polar Great Power*, ch. 2. On the propaganda system itself, Brady, *Marketing Dictatorship*.
[8]Volpe, "China's Arctic Policy."
[9]See Appendix.

insights without compromising interviewee safety or anonymity. *Neican* reports are confidential and never published. A major limitation of this approach is that the analysis reconstructs the mechanism of expert involvement in policy from interviews with experts and bureaucrats and from institutional documents and secondary sources, without access to *neican* texts.

Literature review

Broadly speaking, literature on science–policy interaction identifies two ways of thinking about the role of expertise in policymaking: linear and co-constitutive.[10] Linear models conceptualise knowledge as an input that precedes and informs politics. Expert networks with a claim to policy-relevant knowledge analyse a problem and pass the analysis to decision-makers, who can then act under reduced uncertainty.[11] Zhu's study of expert involvement in Chinese policy change finds that the linear pattern occurs mainly when policymakers lack the expertise to choose among options and where those who stand to lose from the change are not themselves close to decision-makers.[12]

In contrast to linear models, co-constitutive models treat expertise and expert knowledge as politicised. Jasanoff argues that what counts as a policy problem and what counts as relevant expertise are co-produced.[13] This means expertise is an outcome of the policy process, making competition among would-be experts a struggle over position rather than a contest in which the best knowledge wins.[14] This requires experts to adjust their views to dominant expectations, as recognition mediates what counts as authoritative knowledge.[15] However, if authority is conferred through recognition, what matters is on what grounds recognition is sought and who sets the criteria by which it is granted.[16] Kaczmarska observes that knowledge and expertise studies long presumed that knowledge produced under political control is simply corrupt and not useful.[17] Knowledge and policy are co-produced when the categories in which policy problems are posed are jointly constituted by the state and experts, and this goes beyond the state commissioning, funding, and screening expertise.[18] To facilitate co-production, standardised structures are needed that serve both experts and policymakers and provide the context within which particular objects move between them; these are boundary infrastructures.[19] Boundary objects that cross them, such as policy briefs or maps, are flexible enough to serve several communities of practice and 'may be used as tools in knowledge co-production processes'.[20] In practice, the roles available to experts and policymakers on the boundary are blurred, and the same actors may perform several of them within

[10]Glaab and Kortendiek, "The Politics of Knowledge Production," 167–77.
[11]Haas, "Epistemic Communities and International Policy Coordination," 2–3.
[12]Zhu, *The Politics of Expertise in China*, vi–vii, 29–32.
[13]Jasanoff, *States of Knowledge*, 3.
[14]Glaab and Kortendiek, "The Politics of Knowledge Production in International Organizations," 171–77.
[15]Sending, *The Politics of Expertise*, 8.
[16]Ibid., 6–7.
[17]Kaczmarska, "Context-Centred Approaches to Knowledge," 473–74; Kaczmarska, "Academic Freedom and the Contexts of Knowledge Production," 554–55.
[18]Waisová, "Deep Co-production of Human Security at the Science–Politics Nexus," 238–42.
[19]Bowker and Star, *Sorting Things Out*; and Star, "This Is Not a Boundary Object."
[20]Sarkki et al., "How Boundary Objects Help to Perform Roles of Science Arbiter, Honest Broker, and Issue Advocate," 162.

a single process.[21] Boundary organisations are institutions that facilitate the creation of boundary objects and co-production between experts and policymakers.[22]

In China, the party-state signals demand for expertise through the topical composition of funding calls, policy documents on the role of , and leaders' statements, and experts respond to these signals.[23] The signals are often vague and underspecified. For example, signature concepts may signal leadership's policy priorities, but their vagueness allows domestic actors to fill them with meaning, which allows the party-state to accommodate competing interests and secure support.[24] Policymakers also have incentives to withhold key information about decisions even as they solicit advice.[25] Therefore, experts cannot be certain about the need for expertise. However, experts do not simply wait to be summoned; they adopt indirect strategies to remain relevant or pursue policy entrepreneurship.[26]

Empirically establishing the linear influence of expertise on policy is difficult. If an expert framing appears before a matching policy shift, this may indicate expert influence or that the expert correctly anticipated the policy shift. Likewise, the state may have several reasons to solicit advice. Chinese policymakers confident in their own judgement may use expertise to authenticate a decision already taken, explain it publicly, or legitimise the process.[27] This article examines how they are co-produced in a policy consultation regime that relies on anticipating needs under bidirectional information asymmetry. It uses these boundary concepts to distinguish the practices examined below.

Politics of expertise in China

The influence of experts on policy in the People's Republic of China (PRC) is well-documented. In foreign policy, Chinese scholars and think-tank experts actively contribute to shaping and interpreting foreign policy.[28] Broadly speaking, experts can influence policy actors directly or indirectly. They may directly use formal or informal channels of communication with policymakers to provide 'their research results (in oral or written formats) and try to make policymakers understand and accept their policy opinions'.[29] Direct involvement may include writing and submitting reports,[30] participating in expert and advisory committees,[31] and taking part in policy discussions and seminars at various levels, the highest of which being collective study sessions. Indirect involvement entails participation in public discourse by publishing academic articles or undertaking media appearances. However, the specific mechanisms of expert influence on policy remain understudied.

[21]Ibid.
[22]Guston, "Boundary Organizations in Environmental Policy and Science."
[23]Mokry, *Chinese Scholars and Think Tanks' Constructions of China's National Interest*, 23–24.
[24]Zeng, *Slogan Politics*.
[25]Zhu, *The Politics of Expertise in China*, 26–28.
[26]Ye and Xue, "Motivated to Be Policy Entrepreneurs."
[27]Zhu, *The Politics of Expertise in China*, 27–28.
[28]Mokry, *Chinese Scholars and Think Tanks' Constructions of China's National Interest*.
[29]Zhu, *The Politics of Expertise in China*, 19.
[30]Tsai, "A Unique Pattern of Policymaking in China's Authoritarian Regime"; Ye and Xue, "Motivated to Be Policy Entrepreneurs."
[31]Li, Chen, and Wu, "Managing National Expert and Advisory Committees in the Chinese Policy Advisory System."

The decision-making system in the PRC relies on the 'consultative information system', which in turn relies on the collection of policy proposals from institutions outside formal decision-making bodies. The consultative system, known as 'work to inform party and government organs' (*dangzheng jiguan xinxi gongzuo*), acquired its current form in the 1980s.[32] Though the system originally primarily involved leading Chinese media outlets like the *Xinhua News Agency* and *The People's Daily*,[33] this system was expanded to encompass academic institutions, including think tanks and universities.[34] This has especially accelerated since the 2010s, with the decision to promote a 'new type of think tanks'.[35] Much recent literature has focused on expert involvement in policymaking through these formal and informal consultation channels.

Specifically, Chinese experts provide policy advice by writing advisory reports (*zixun baogao, zhuanbao*), which are compiled and disseminated through CCP channels in the form of *neican* ('internal references', *neibu cankao*).[36] Scholars write reports and submit them to party organs, which decide whether to accept them for publication.[37] Leading institutions and think tanks have quantitative requirements for the writing of advisory reports. The General Office of the CCP Central Committee has the final authority on compiling these reference collections and forwards the most relevant reports to the highest party leadership for review.[38] After reading the reports, party leadership of various levels write instructions (*pishi*), usually in the form of notes on the margins of the report, providing specific instructions to lower level organs. Receiving a *pishi* is a relatively rare event and a testament to the quality of the report, confirming the importance of the issues raised by the expert in it.[39] Experts' ability to attract *pishi* can be considered a special form of political capital enabling them to attract attention and resources.[40] The opaque internal correspondence system operates through the party rather than the state apparatus and is increasingly the most effective way for experts to feed into the policymaking process. Though similar mechanisms of expert involvement exist in other countries, the scale and degree of institutionalisation is unique to China. *Neican* are usually classified depending on the level of the official who read them.[41] The *neican/pishi* system plays an important role in evaluating Belt and Road Initiative projects, for example.[42]

[32]Liu, "A Brief Account of the Development of Information Work."
[33]Tsai, "A Unique Pattern of Policymaking in China's Authoritarian Regime."
[34]Abb, "China's Foreign Policy Think Tanks"; Shambaugh, "China's International Relations Think Tanks"; and Zhu, "The Influence of Think Tanks in the Chinese Policy Process."
[35]Hayward, "The Rise of China's New-Type Think Tanks."
[36]Abb, "China's Foreign Policy Think Tanks"; Tsai, "A Unique Pattern of Policymaking in China's Authoritarian Regime"; Tsai and Liao, "Concentrating Power to Accomplish Big Things"; Wang, Liu, and Chen, "Cooperation and Conflict Studies in Chinese IR"; Ye and Xue, "Motivated to Be Policy Entrepreneurs"; and Zhao, "Dig Deep and Reach Wide."
[37]Ye and Xue, "Motivated to Be Policy Entrepreneurs."
[38]Tsai, "A Unique Pattern of Policymaking in China's Authoritarian Regime"; and Zhao, "Dig Deep and Reach Wide."
[39]Tsai, Li, and Song, "Agent Politics of Chinese Think Tanks"; and Zhu, "The Influence of Think Tanks in the Chinese Policy Process."
[40]Zhao, "Dig Deep and Reach Wide."
[41]Qingzu ai shuo yule, "What Level of Authority Allows Access to Internal Documents in My Country?"
[42]Li and Wang, "Adaptive Governance at Work."

What counts as polar expertise in China

Polar expertise is an ad hoc analytical category. There is no institutionalised term for polar expertise in China that certifies anyone as a polar expert *tout court*. Polar science in China can be broadly divided into hard (*ying kexue*) and soft sciences (*ruan kexue*). The former are natural and technological sciences (biology, climatology, oceanography, physics, glaciology, marine sciences, geology, etc.). The latter are social sciences and humanities, whose research areas cover 'strategy, decision-making, planning, policy, science and technology legislation, technology forecasting, technology evaluation, management science, science of science, technology and economic analysis, feasibility studies, and consulting techniques' etc.[43] In addition to science, technical experts, such as shipbuilding, shipping and petroleum industry experts, military strategists, as well as bureaucrats and private consultants, possess expertise about the polar regions.

Under the twelfth Five-Year Plan, the State Oceanic Administration (SOA) launched the Comprehensive Investigation and Assessment of the Polar Environment programme, overseen by a joint SOA–Ministry of Finance leading group. This hard-science–dominated programme included soft science for 'strategic assessment of national interests in the polar regions', tasked with researching polar geopolitics, resources, international law, and national polar policies.[44] These soft science outputs were subject to formalised quality and security controls.[45] In 2010, the Polar Research Institute of China (PRIC) received SOA funding to expand social science and establish a Division of Strategic Studies to research selected strategic themes.[46]

Both hard and soft sciences in China are supported 'only to the extent that the professional knowledge serves the state's utilitarian purposes'.[47] However, the party-state treats hard science output in itself as an instrument to enhance its performance legitimacy and international status and thus has higher tolerance for hard science without immediate technological applications, provided it yields high publication counts.[48] In polar science, possession of scientific equipment and producing science become a measure of the technological power of the state.[49] In the polar domain, soft sciences depend on policy usefulness. Fields like anthropology and sociology remain under-represented because funding and evaluation incentives favour international relations and law. The Ministry of Science and Technology's funding guidelines for its National Key Research and Development Programmes, for example, solicited research on polar law regarding Arctic shipping routes, and the 2019 guidelines set a target of establishing 'international polar cooperation mechanisms with no fewer than two countries/regions'.-[50] Thus, Chinese experts retain some scientific autonomy but are also politically engaged.

Polar expertise is organised by discipline and institution. The consultation regime is not designed to transmit findings linearly from a defined body of experts but to provide

[43]Li and Pan, "Zhongguo Nanji Ruankexue Yanjiu de Yiyi, Xianzhuang Yu Zhanwang," 224.
[44]*Zhongguo haiyang nianjian 2012*, 449; and Wang, *Intentions and Unintended Consequences*, 16.
[45]*Zhongguo haiyang nianjian 2015*, 430.
[46]Wang, *Intentions and Unintended Consequences*, 213.
[47]Gu, "The Co-Production of Normal Science," 98.
[48]Ibid.
[49]Lamazhapov, "Polar Regions for Global Status"; and Kossa, "China and Arctic Science in the Xi Jinping Era," 238–240.
[50]Eiterjord, "Polar Expertise in China's 14th Five-Year Plan."

practices for acting under mutual uncertainty. In the following sections, the article first explains how expertise reaches policymakers, then examines experts' institutional positions. Finally, it discusses three practices: horizontal bodies coordinating expertise across *xitong*, ad hoc expertise mobilisation for a major policy document, and routine internal reports produced without commission.

Institutional arrangement of polar policy institutions in China

The existing literature has mapped the institutional layout of China's polar apparatus.[51] The section below describes it also in terms of epistemic authority. Most experts are subordinate to state institutions under the State Council and its 26 constituent departments or provincial governments. In addition, the State Council directly supervises multiple institutions at the ministerial level, including provinces, as well as others. Most of the ministerial-level organisations under the State Council manage their own vertically integrated departmental systems (*xitong*).[52] In Chinese polar policymaking, the Ministry of Natural Resources (MNR), Ministry of Foreign Affairs (MFA), Ministry of Transport, Ministry of Science and Technology, Ministry of Education, and Ministry of State Security are the most important *xitong*. The Chinese Academy of Sciences, the Chinese Academy of Social Sciences, and provincial governments are also under the State Council at the ministerial level. *Xitong* institutional arrangements that experts operate within may refer both to formal administrative structures (such as the MFA *xitong*, with MFA at the top) and decision-making systems (*foreign policy xitong*).[53] If one takes foreign policy as a whole, it is the Commission of the CCP Central Committee for Foreign Affairs that is the top authority, not the MFA.[54]

From an organisational perspective, the MNR 'maritime' *xitong*, which includes the now-defunct SOA, is the competent authority for polar affairs in China.[55] The SOA exists in name only (headed by an MNR vice-minister), but until 2018 it oversaw all the country's marine affairs, including polar expeditions, before it was merged, along with the State Bureau of Surveying and Mapping, into the Ministry of Land and Resources, which became the MNR. The Chinese Arctic and Antarctic Administration (CAA) was (and still is on paper) subordinate to the SOA but now functions as 'one institution with two nameplates' (*yige jigou liang kuai paizi*). Notable polar policy institutes within the MNR system include the PRIC, the China Institute for Marine Affairs, and the First, Second, and Third Institutes of Oceanography. The main financing document for polar research, the *National Five-Year Plan for Polar Expedition and Development*, is 'developed mainly by SOA (or MNR) with very limited input from the general scientific community'.[56]

The 2018 restructuring did not resolve who in the MNR was responsible for the polar portfolio. While the CAA's Division of International Cooperation traditionally handled

[51]Brady, *China as a Polar Great Power*; Kobzeva, "China's Arctic Policy"; Kossa, "The Arctic in China's National Strategy"; Tonami, "The Arctic Policy of China and Japan"; and Zhang et al., "Reforming China's Polar Science and Technology System."
[52]Chen, "China's Emerging Arctic Strategy."
[53]Brady, *China as a Polar Great Power*, 137; and Lanteigne, *Chinese Foreign Policy*, 37.
[54]Wang, "Zhongguo Canyu Beiji Zhili de Duoyuan Zhuti Xietong Jizhi Yanjiu," 60.
[55]Brady, *China as a Polar Great Power*, 119–21.
[56]Zhang et al., "Reforming China's Polar Science and Technology System," 390.

international polar cooperation, the MNR Department of International Cooperation also has authority over certain initiatives, and the PRIC engages in significant international activity. Wang has documented how this uncertainty over mandates and intra-institutional competition limited ministerial support for the initiative in question.[57] Crucially, this competition concerned which unit is credited for achievements within the MNR. Fragmentation of authority in polar affairs is also present below the ministry level.

MFA is an external-facing system. The Department of Treaty and Law at the MFA oversees polar policy, and China's special representative for the Arctic is appointed from there.[58] The Department of Treaty and Law is tasked with 'coordinating the compliance of international treaties' and 'organises the participation in diplomatic negotiations on climate change and environmental treaties',[59] which includes the United Nations Convention on the Law of the Sea and the Antarctic Treaty. The department is an organ within the MFA with specialised expertise in international law.

Provincial governments also play a significant role in policymaking in China.[60] In China's polar policies, arguably the most notable provincial government is Shanghai, which is the homebase of the PRIC and thus of China's polar expeditions as well as of other important institutes, universities, and think tanks. Several other provinces also have important roles.[61] Departments of Natural Resources in several provinces also operate provincial oceanic administrations, with more of them opening recently.[62] Provincial expert organs are under 'double leadership', answering both to provincial governments geographically (*kuai*) and to the central government's functional agencies vertically (*tiao*).[63]

Chinese research institutions and think tanks are considered sources of expertise that the party-state can utilise in policymaking and public diplomacy.[64] Outside the MNR *xitong*, most polar science is produced in other research institutes and universities. Universities are usually subordinate to the Ministry of Education, though there are some exceptions.[65] Dalian Maritime University, for example, is part of the Ministry of Transport. Some universities are under provincial governments. Academies of sciences, like Chinese Academy of Sciences, manage their own constituent institutes. Some state-owned enterprises also manage institutes, like the 708 Research Institute of the China State Shipbuilding Corporation. Think tanks may be a part of different *xitong*.

A public institution's (*shiye danwei*) position affects its autonomy in addition to its reporting. Since the 2011 State Council reform, they are classified into fully state-funded Category I institutions with strict oversight, and partially funded Category II institutions

[57]Wang, *Intentions and Unintended Consequences*, 281, 286–87.
[58]Gao Feng was China's special representative for the Arctic between 2016 and 2024. There is no special representative at the time of writing. Interview with bureaucrat B, April 2025. See also Kossa, *The Arctic in China's National Strategy*, 36–37.
[59]Ministry of Foreign Affairs of the People's Republic of China, "The Department of Treaty and Law."
[60]Kossa, *The Arctic in China's National Strategy*, 36–37.
[61]Kossa, "China's Arctic Engagement."
[62]Liaoning Province, "The Nameplate of the Liaoning Province Oceanic Administration Officially Hung Up."
[63]Ma, "Vertical Environmental Management."
[64]Kossa, *The Arctic in China's National Strategy*, 36–37.
[65]Tongji University, Wuhan University, Ocean University of China, Shanghai Jiao Tong University, Dalian Maritime University, Sun Yat-sen University, Xiamen University, Jilin University, Fudan University, and Beijing Normal University are some of the universities with the most polar expertise.

that may use market mechanisms for resource allocation and have greater autonomy. The PRIC is a department-rank Category I institution directly under the MNR, and its website states that it upholds and strengthens the party's overall leadership in carrying out its mission. Likewise, the Chinese Academy of Sciences and the Chinese Academy of Social Sciences, both ministry-rank bodies directly under the State Council, describe themselves first and foremost as political institutions. The practical salience of such self-descriptions varies between organisations and over time, but the classification ensures political functions take precedence over scientific ones when conflicts arise. Party structures often form a parallel pillar (*tiao*), mirroring and permeating all state institutions. For example, the General Office of the Central Committee of the CCP often plays a more central role than the State Council, which has weakened in recent years.[66] Party groups (*dangzu*) make ultimate decisions about significant appointments, major investment projects, and large expenditures (*sanzhong yida* issues) at all levels,[67] though the same organisations or people often perform these functions. Famously, the State Council Information Office is just an external nameplate for the CCP's Central Propaganda Department.[68]

The institutional setup determines ministries' mandates and priorities. The maritime *xitong* focuses on polar science and resource assessment; the MFA on treaty compliance and diplomacy; the Ministry of Transport on shipping; and the security *xitong* on military issues. These priorities shape knowledge production, as experts work within institutions belonging to specific *xitong*. Institutes have distinct funding, data, and reporting priorities, influencing their advice. Fragmented authority in polar affairs reflects both decision-making structures and limits of knowledge production.

Practices of expert involvement

Horizontal advisory bodies

There are several formal channels of expertise in China. In the maritime *xitong*, there are two main 'multi-stakeholder advisory bodies' on polar affairs.[69] These bodies are horizontal and meant to coordinate work across *xitong* for bodies of roughly the same level (Figure 1). The first is the Chinese Advisory Committee for Polar Research (CACPR, *Zhongguo jidi kaocha gongzuo zixun weiyuanhui*), which is an advisory body to the department in charge of polar expedition work established in 1994 alongside CAA.[70] Its director is the serving deputy director of the SOA, and its secretariat sits within the CAA.[71] CACPR's main responsibilities are to provide consultation to relevant ministries

[66]Tsai and Liao, "The Authority, Functions, and Political Intrigues of the General Office of the Chinese Communist Party."
[67]Beck and Brødsgaard, "Corporate Governance with Chinese Characteristics"; and Ma, "Vertical Environmental Management."
[68]Brady, *Marketing Dictatorship*, 13; and Lulu, Jirouš, and Lee, *Xi's Centralisation of External Propaganda: SCIO and the Central Propaganda Department*.
[69]Wang, "Zhongguo Canyu Beiji Zhili de Duoyuan Zhuti Xietong Jizhi Yanjiu," 56.
[70]Chaired by the SOA, 13 ministry-level institutions send experts to the CACPR: MFA, National Development and Reform Commission, Ministry of Education, Ministry of Science and Technology, Ministry of Industry and Information Technology, Ministry of Finance, Ministry of Land and Resources, Ministry of Agriculture, Ministry of Health, Chinese Academy of Sciences, China Meteorological Administration, National Natural Science Foundation of China, and State Bureau of Surveying and Mapping (see Tonami, "The Arctic Policy of China and Japan'). CACPR does not seem to be active following SOA's abolition in 2018.
[71]Zhang et al., "Reforming China's Polar Science and Technology System," 391.

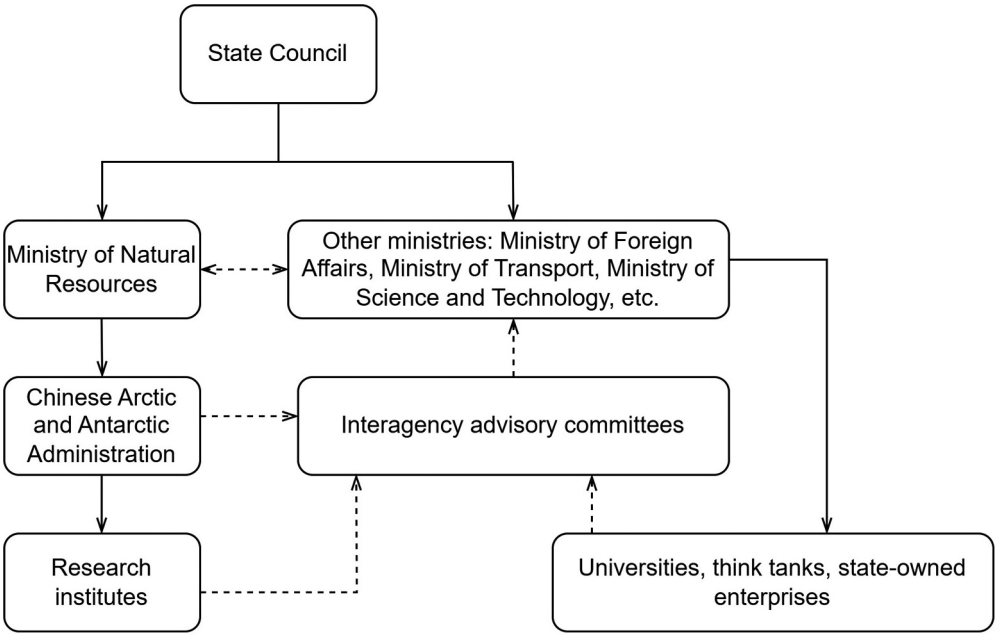


Figure 1 Interagency advisory committees as horizontal coordination organs.

on polar research, organise academic exchanges, provide scientific and technological consultation on polar research, evaluate scientific and technological achievements, and guide scientific research activities.[72] There have been no reports on CACPR's activities since its seventeenth meeting in 2016.[73]

In 2017, China established the Chinese Polar Science and Technology Commission (CPSTC, *Zhongguo jidi kexue jishu weiyuanhui*), which consists of 41 members, including academicians of the Chinese Academy of Sciences and the Chinese Academy of Engineering, university professors, experts from research institutes and related engineering units,[74] as well as experts on science, education, international relations, and economy.[75] Former Minister of Science and Technology Xu Guanhua heads CPSTC. According to the then-head of the SOA, CPSTC's role is to provide advice and suggestions for 'the country's major polar development strategies', 'conduct consultations on major national polar projects', and advise on managing 'research resources'.[76] It is not clear how active CPSTC is because not much has been published about it after its establishment, though multiple members list it on their CVs. This body also coordinates horizontal cooperation among dispersed expert organisations.

The State Council coordinates the Interagency Coordinating Mechanism on Arctic Affairs as one of the main advisory bodies under the Department of Treaty and Law of

[72] *Zhongguo haiyang nianjian 1994–1996*, 1997, 269.
[73] Wu, "Zhongguo Jidi Kaocha Gongzuo Zixun Weiyuanhui Di 17 ci Huiyi Zai Shanghai Zhaokai."
[74] Liu, "Zhongguo Jidi Kexue Jishu Weiyuanhui Chengli."
[75] University of Science and Technology of China, "Shiying Xin Shidai Haiyang Qiangguo Zhanlue Xuqiu: Wo Xiao Jiji Canyu Guojia Jidi Keji Fazhan Yanjiu."
[76] Liu, "Zhongguo Jidi Kexue Jishu Weiyuanhui Chengli."

the MFA.[77] It was established in 2011 and was headed jointly by the MFA and the SOA.[78] It is a 'loose coordination mechanism, whose main tasks were to share information and assess the situation in the Arctic' and to coordinate the work of different ministries and improve information sharing.[79] While the MFA has a coordinating function, its decision-making power is constrained not only by other functional ministries but, more crucially, by CCP Leading Small Groups.[80] From 2012 until 2018, polar affairs fell under the purview of the Central Leading Small Group for Safeguarding Maritime Rights and Interests, which was dissolved as part of the 2018 reforms and merged into the upgraded Central Foreign Affairs Commission and its office.[81] Some Chinese commentators worried that this move weakened the overall prominence of maritime affairs at the highest level.[82] No dedicated body oversees polar work at the leadership level, and operational coordination falls back to functional ministries. For example, Wang shows that responsibility for the China–Iceland Arctic Science Observatory was contested between the MFA and the MNR.[83] Within the MFA, the only dedicated diplomatic position is the Special Representative for Arctic Affairs established in 2016. However, even this position has been vacant since late 2024. These changes suggest that the overall salience of Arctic affairs at the leadership level is lower than often assumed.

These bodies were created to coordinate horizontally across ministries on complex problems. They can also be understood as boundary organisations, since they link expertise and bureaucracies. Their declining prominence suggests that horizontal coordination has partly moved towards other associations and learned societies. The Polar Science Branch of the Chinese Society for Oceanography and the Polar Branch of China Ocean Engineering Consulting Association primarily focus on polar science and engineering, linking the PRIC and CAA with other scientists, port engineering companies, heavy machinery manufacturers, aviation manufacturers, and applied research institutes, as well as providing consulting services for polar strategies and planning.[84] In contrast, the Polar Development Branch of the China Association of Marine Affairs, established in June 2018 under the MNR, 'submits recommendations on polar legislation, policies, plans, and systems to relevant national authorities', and brings together MNR-affiliated research institutes, universities, and think tanks.[85] Finally, the Polar Navigation and Equipment Professional Committee of the China Institute of Navigation gathers polar shipping actors including COSCO, Jiangnan Shipyard, and the CSSC 708 Research Institute to advise on icebreaker and high ice-class vessel construction.[86]

Regular annual conferences and forums have taken on some of these roles. Chief among them is the annual China Symposium on Polar Science (*Zhongguo jidi kexue xueshu nianhui*), convened since 2005 by the PRIC with the Chinese Society for

[77]Kobzeva, "China's Arctic Policy."
[78]Jia, "Speech by Deputy Director-General Jia Guide at the Inaugural Ceremony of the CNARC."
[79]Jia and Shi, "Dui Xin Xingshi Xia Zhongguo Canyu Beiji Shiwu de Sikao," 24–25.
[80]Lai and Kang, "Domestic Bureaucratic Politics and Chinese Foreign Policy."
[81]Wang, "Zhongguo Canyu Beiji Zhili de Duoyuan Zhuti Xietong Jizhi Yanjiu," 60.
[82]Yang, "Haiyang jibenfa de lifa dingwei yu tixi jiegou," 123.
[83]Wang, *Intentions and Unintended Consequences*, 166.
[84]PRIC, "Zhongguo Haiyang Xuehui Jidi Kexue Fenhui"; and PRIC, "Zhongguo Haiyang Gongcheng Zixun Xiehui Jidi Fenhui."
[85]PRIC, "Zhongguo Haiyang Fazhan Yanjiuhui Jidi Fazhan Fenhui."
[86]PRIC, "Zhongguo Hanghai Xuehui Jidi Hangxing Yu Zhuangbei Zhuanye Weiyuanhui"; and China Institute of Navigation, "Zhongguo Hanghai Xuehui Jidi Hangxing Yu Zhuangbei Zhuanye Weiyuanhui."

Oceanography and hosted on a rotating basis by universities and municipal governments. Its stated purpose is to build bridges for cross-disciplinary, cross-departmental, and cross-institutional research collaboration. For example, the 2026 symposium aimed to 'actively formulate China's action plan for the Fifth International Polar Year (IPY-5)'.[87] Polar policy and law is one of fifteen sub-forums.

Conferences like this are flexible and assemble different constellations of research organisations, ministerial and provincial actors, and state-owned enterprises. A survey of 95 domestic and international Arctic social science conferences held between 2007 and 2022 shows that the former SOA attended 27 times and the MNR 14 times, followed by the China Classification Society at eight, the MFA at seven, and the Maritime Safety Administration at four.[88] COSCO Shipping and its subsidiaries attended 19, and China National Petroleum Corporation attended two.[89] These bodies do not openly include organisations from security and defence *xitong*. Security and military analysts produce polar security knowledge, but the extent of their cooperation with civilian polar science and advisory bodies is unclear. Security expertise seems to remain largely isolated from other expert communities.[90]

Ad hoc mobilisation: drafting the 2018 Arctic policy white paper

The process of writing of China's Arctic Policy White Paper, which was published at a joint press conference of the State Council Information Office and the MFA in 2018, exemplifies the ad hoc mobilisation of expertise, and Chinese experts contributed to writing the policy.[91] As early as 2011, Chinese scholars gave policy advice that China should 'introduce China's Arctic policy'.[92] Many voices have raised the argument that China should 'formulate China's Arctic policies and countermeasures from the perspective of safeguarding China's Arctic rights and interests'[93] and fretted that 'China has not yet issued its own Arctic strategy or official Arctic policy documents, which to some extent has led to speculation and conjecture about the purpose of China's participation in Arctic affairs'.[94] The process gained particular traction after China became an observer in the Arctic Council in 2013.

In 2014, policymakers came to the same conclusion. Two officials at the MFA's Department of Treaty and Law wrote that there was a need to write 'a White Paper on our country's [China's] Arctic policy'.[95] The goal of such a White Paper would be to 'provide correct guidance for Arctic-related work domestically' and thus to coordinate the 'efforts of relevant units in carrying out Arctic affairs', as well as to 'properly clarify' China's position internationally and 'dispel doubts'.[96] Clarification of China's position

[87]PRIC, "2026 Zhongguo Jidi Kexue Xueshu Nianhui Zai Nantong Zhaokai."
[88]Li and Li, "Beiji Yanjiu Guonei Zhuti Yantaohui de Yiti Jiegou ji Fazhan Qushi," 77.
[89]Ibid.
[90]A notable example of cooperation is the MSS-affiliated Jiangnan Social University Professor Lu Junyuan, who partook in SOA's Comprehensive Investigation and Assessment of the Polar Environment. Lu and Zhang, *Zhongguo beiji quanyi yu zhengce yanjiu*.
[91]Kobzeva, "China's Arctic Policy: Present and Future."
[92]Xia, "Beiji Huanjing Bianhua Dui Quanqiu Anquan He Zhongguo Guojia Anquan de Yingxiang," 133.
[93]He and Zhang, "Cong Beiji Guojia de Beiji Zhengce Pouxi Beiji Keji Fazhan Qushi," 413.
[94]Sun and Zhang, "Beiji Bianqian Shijiao Xia Zhongguo Beiji Liyi Gongtongti de Goujian," 96.
[95]Jia and Shi, "Dui Xin Xingshi Xia Zhongguo Canyu Beiji Shiwu de Sikao," 27.
[96]Ibid.

was felt to be all the more important as the MFA officials bemoaned that rogue 'experts, scholars, and officials in China have published erroneous views and statements' about China's position on the Arctic, such as '"the Arctic belongs to all mankind",' 'which have exacerbated the misunderstandings and concerns of Arctic countries'.[97] Writing a properly articulated document was seen as a solution that could signal China's clear position to both domestic and international actors.

Wang Chao, who was the Vice-Minister of Foreign Affairs between 2013 and 2019, played a crucial role in the 'progress and advancement' of the White Paper, based on the 'appeals' (*huyu*) made by scholars, because all the other Asian states 'like Korea, Japan, and even Singapore all had Arctic discourses' (*huayu*), so 'China also needed to have one'.[98] China's Arctic Policy was written in close cooperation with researchers. This is because 'expertise is a prerequisite for participation in governance'.[99] The MFA commissioned specific scholars to write reports and suggestions for the White Paper on China's Arctic Policy. Several institutions participated directly in its writing. As scholar L said,

> I have personally not participated in writing the White Paper, but I've heard from others that when the policy was written, maybe a small group of scholars (*yipi zhuanjia*) was gathered in Beijing, especially scholars from Qingdao, very influential scholars. And they wrote a series of reports that went into the forming of the White Paper.[100]

Many scholars contributed to drafts of the White Paper and made numerous proposals for different texts that could be included in it.[101] A concrete example is the term 'near-Arctic state' (*jin beiji guojia*) that scholars Zhang Xia and Lu Junyuan proposed and had used in their previous publications.[102] The term was included in the final version of the White Paper, thus becoming codified into China's official policy.[103]

Different factors, such as scholars' institutions and personal connections, play a role in being invited to contribute as an expert. For example, the status of the Shanghai Institutes for International Studies (SIIS) within the MFA *xitong* rose after Yang Jiemian became SIIS director in 2007. Yang Jiemian is the brother of Yang Jiechi, who was the Minister of Foreign Affairs in 2007–2013 and the Director of the Office of the Central Foreign Affairs Leading Group (later Commission) in 2013–2022. It was during Yang's tenure as director in July 2008 that SIIS's name was changed from 'Institute' (*yanjiusuo*) to 'Institute*s*' (*yanjiuyuan*, 'academy' in Chinese). Scholars from leading think tanks like SIIS were even temporarily transferred to work at the Department of Treaty and Law of the MFA.[104]

Scholars are rather hesitant about their overall influence on policy, though. In the grand scheme of things, a situation like drafting the Arctic Policy White Paper is not usual. According to Zhu's schemata, this is a situation where the state's knowledge about the region is low, and thus scholars are needed.[105] As bureaucrat B put it, 'we say that

[97]Ibid., 26.
[98]Interview with scholar J, April 2025.
[99]Interview with scholar E, April 2025.
[100]Interview with scholar L, April 2025.
[101]Interview with scholars H, J, K, L, April 2025.
[102]Lu, *Beiji Diyuan Zhengzhi Yu Zhongguo Yingdui*, 339; and Lu and Zhang, *Zhongguo beiji quanyi yu zhengce yanjiu*, 380.
[103]See also Kossa, *The Arctic in China's National Strategy*, 75; and Mokry, *Chinese Scholars and Think Tanks' Constructions of China's National Interest*, 1.
[104]Zhao, interview by *Pengpai*, August 13, 2025.
[105]Zhu, *The Politics of Expertise in China*, 27.

China is a near-Arctic state. It's kind of like if you were born in the desert, and you've never seen a river, you wouldn't know how to swim. So we have to learn, such as from Nordic countries'.[106] Indeed, before 2010, China did not possess anything like the same level of expert knowledge about the region, which resulted in many significant scientific projects being funded. As scholar M stated,

> The state kicks the ball rolling and indicates the general direction. This can be achieved with the help of different instruments, such as funding or speeches. The role of the state is in a way to open the door for others to start forming policy . . . . The entrepreneurs on the lower level, whether it's universities or centres, may become research entrepreneurs and take advantage of these opportunities given by the state, and then it becomes a wide-open race that allows them to build up their institutional strength by setting up new schools or attracting academics.[107]

The White Paper also functions as a boundary object. For the party-state, it both guides domestic Arctic work and clarifies China's stance internationally. The White Paper was co-produced with the expert community because the leadership acknowledged its knowledge and coordination needs. The state set the broad political objective and selected acceptable advice, while experts had already developed relevant concepts and framings and helped define its content, with 'near-Arctic state' as a specific contribution from scholarly work. The White Paper was partly motivated by status parity with Asian states. Co-production can also serve to discipline expertise, as it was meant to correct what was seen as erroneous statements.

Routine policymaking through *neican/pishi*

Experts are less likely to be involved in policy advisory once there is sufficient knowledge about a particular policy area within State Council organisations. As scholar G put it, the role of experts is in 'raising the state's and the government's level of attention to the Arctic issues'.[108] The internal reporting system (Figure 2) becomes the main mechanism for expert involvement. In recent years, the importance of utilising the internal report system has increased, and many more scholars take advantage of the internal party reporting to become policy entrepreneurs. This is also due to the increased role that the Central Party Organisation plays in policymaking, where there is more policy advice directly towards the party centre.[109] As bureaucrat B put it,

> There are two highest policy organisations in China, the General Office of the CCP and the Development Research Centre of the State Council. Because the MFA implements whatever policy the General Office tells them to. You can say that the General Office reports directly to the personal secretary of Xi Jinping. So, it has much more freedom than the MFA.[110]

[106]Interview with bureaucrat B, April 2025.
[107]Interview with scholar M, April 2025.
[108]Interview with scholar G, April 2025.
[109]Interview with scholar M, April 2025.
[110]Interview with bureaucrat B, April 2025.

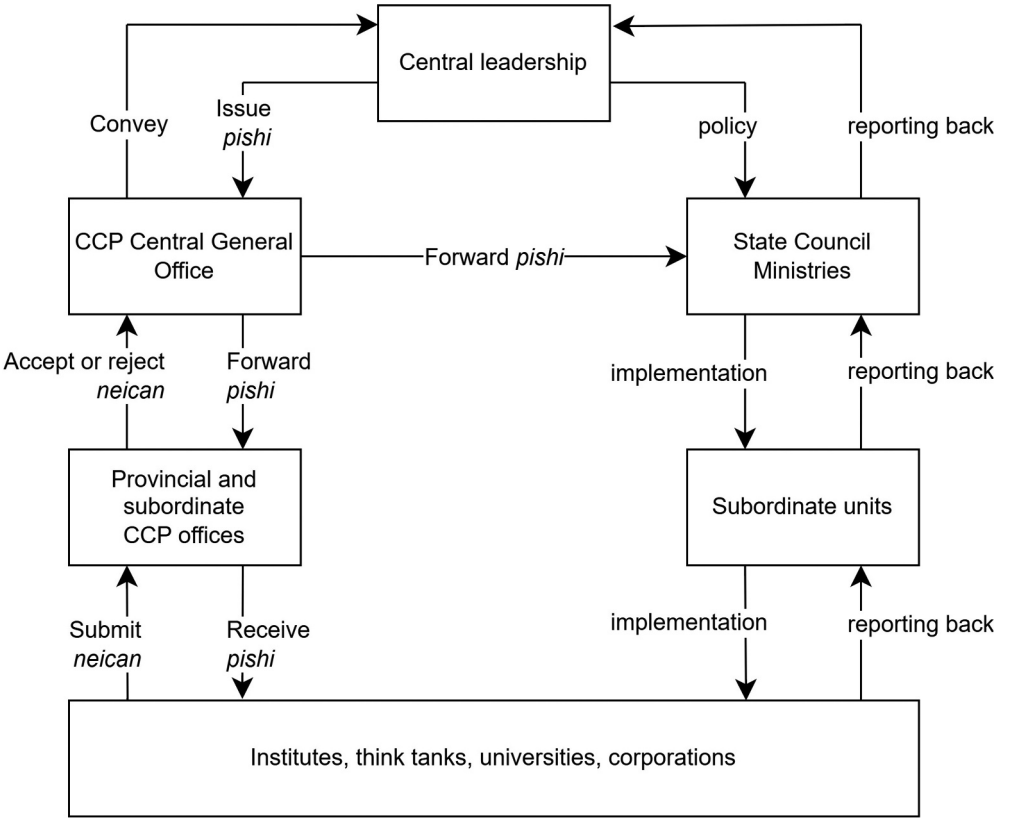


Figure 2 Structure of the *neican/pishi* reporting system. on the left side are party institutions, on the right are the state institutions.

The internal information system exists in parallel with the formal system of public administration.[111] It is somewhat at odds with vertically integrated departmental systems (*xitong*). Formally, institutes, universities, and think tanks may belong to the same departments but report through other party organs to the internal information system. To a certain extent, an overlap exists in responsibilities between the expanded internal reporting system and the old *xitong* system. As Zhao comments, one reason the *neican* system has been expanded is that information gathered through traditional *xitong* channels is seen as potentially biased towards ministerial interests.[112] Reports can be submitted through different structures; they can 'go through the Ministry of Foreign Affairs, the Ministry of Education, some military structures, but it can be said that this all has been consolidated to serve the party centre'.[113] It seems that there are some quality

[111]Meng Tianguang, "Guojia de Shenjing Xitong." Meng reads the system as the information 'nervous system' of the state, independent of the ordinary administrative hierarchy, addressing both vertical information asymmetry between the centre and localities and the horizontal problem of reading a complex society; he also notes the recent think-tankisation (*zhikuhua*) of the channel, in which think-tank reports become a principal content source for *neican* and *neican* in turn becomes a form in which think-tank output is presented.
[112]Zhao, "Dig Deep and Reach Wide," 61.
[113]Interview with scholar M, April 2025.

controls for the *neican* and that providing accurate information based on contact with foreign experts is encouraged.

Crucially, academics are not the only actors contributing with policy-relevant work, as certain CCP units like the Liaison Department have been expanding into new functional areas and doing more policy-relevant research.[114] The key difference, however, is that bureaucrats have to write reports 'for superiors' (*dui shangmian*) quickly, 'unlike academics that can take a month or two', so bureaucrats have to rely on lower quality sources like news media.[115] The situation differs depending on the type of think tank or research institute. In the words of scholar H, 'We're not publishing a lot [academically], just writing reports for superiors, so it's not easy to distinguish whether we're academics or officials'.[116] Wübbeke identifies the same blurring in the Chinese climate-change expert community, where the boundary between expert and official roles is indistinct in both institutional position and self-understanding.[117]

Besides the bureaucracy, state-owned enterprises are involved in policy work. For example, during Xi Jinping's visit to Russia in 2019, Russia and China agreed to 'more actively utilise the potential' of the Northern Sea Route.[118] Xi Jinping stated that 'China encourages relevant Chinese enterprises to actively participate' in Arctic shipping.[119] A 2019 marked the height of discussions about the Polar Silk Road in China. Following signals about the Polar Silk Road from the centre, the Shandong Provincial Party Committee and Provincial Government 'issued special *pishi* regarding the development and utilisation of the Arctic shipping route', which was directed to the Provincial Department of Transportation and other departments.[120] The Provincial Department of Transportation, in turn, enlisted Shandong Port Group and COSCO Shipping Specialized Carriers to conduct a specialised study, which resulted in a pilot journey by COSCO vessel *Tian Xi*, which carried paper pulp from Finland to Qingdao.[121] This is an example of the state's policy signalling picked up by provincial party-state organs and sent downward to commission an applied study.

Without seeing the contents of *neican* reports, it is difficult to ascertain whether the language varies significantly from that of public discourse. Traditionally, *neican* report on issues that, for some reason, cannot be discussed publicly.[122] Chinese scholars recommend that 'given the relatively sensitive nature of Arctic issues, consideration should be given to increasing the frequency of closed-door or confidential meetings ... particularly when it touches upon strategy and security'.[123] Experts may feel freer to raise sensitive topics, though other considerations, such as avoiding making trouble, may also lead to self-censorship.[124] The incentive structure discourages the transmission of

[114]Ibid.
[115]Interview with bureaucrat B, April 2025.
[116]Interview with scholar H, April 2025.
[117]Wübbeke, "China's Climate Change Expert Community," 731.
[118]Kremlin, "Zayavleniya dlya pressy po itogam rossiysko-kitayskikh peregovorov."
[119]Ministry of Foreign Affairs of the People's Republic of China, "Xi Jinping Jieshou Eluosi Meiti Lianhe Caifang."
[120]Shandong Provincial Department of Transportation, "Zhongyuan Haiyun Teyun – Shandong Sheng Gangkou Jituan Beiji Hangxian Chenggong Shou Hang."
[121]Ibid.
[122]Chen, " Labor Protests in China in the 1950s"; Li and Wang, "Adaptive Governance at Work"; and Tsai, "A Unique Pattern of Policymaking in China's Authoritarian Regime."
[123]Li and Li, "Beiji Yanjiu Guonei Zhuti Yantaohui de Yiti Jiegou ji Fazhan Qushi," 78.
[124]Li and Wang, "Adaptive Governance at Work."

adverse information upward. Wang Yue documents that when cooperation with Western scientists stalled, actors involved had reason to downplay or conceal adverse signals when reporting within the system.[125] Because advancing a view is combined with retrospective accountability, silence and passive compliance are common as well. As one of Wang's interviewees put it, doing nothing carries no risk of 'backfire'.[126] Furthermore, in the IR field, many regional studies systems have long been under censorship controls. Scholar C reported, for example, that researchers working on the DPRK write *neican* as their primary research output because academic publications on the DPRK are often censored. Therefore, writing *neican* allows researchers to publish their results, despite a certain degree of secrecy around them.

In terms of topics, 'There are two types of topics, the first one is the topics that haven't been paid attention to that someone should pay attention to; the other one are the topics that Xi Jinping wants to know about and might give directions, and we'll write about that'.[127] The criteria by which *neican* are selected are not public. Research funding is normally allocated based on whether project proposals meet 'the objectives and the needs of the country's strategic interests', as well as on their scientific and societal impact.[128] Reports are likely screened on comparable grounds. Anticipation of superiors' preferences thus operates both in the selection of topics and in the handling of unwelcome findings.

The writing of *neican* is an awkward topic to discuss with a foreign academic (like the author) or to explore academically. For example, scholar Q admitted that they write *neican* but did not wish to discuss the topic in detail.[129] Nevertheless, though the text itself cannot be shared, the 'key points' are usually not sensitive.[130]

Thus, in considering *neican*, scholars should not automatically assume that they reveal the hidden intentions or hidden agenda of the PRC or the CCP. From what is known about the system, it is a different and curated register of the Chinese political discourse whose contents partly overlap with public political discourse. Generally speaking, the higher the classification of *neican*, the less ideological distortion, but this does not make its contents more truthful, as authors may still present false, selective, or misinterpreted information.[131]

Indeed, academic articles on polar issues have been reprinted in internal reference publications like *Neibu Canyue* in full.[132] As scholar F said, 'MFA may consult academic papers when making policy'.[133]

Scholars are also very uncertain about the effects that their advice has on policy. For example, scholar F believes that 'in the case of Russia, the influence of academics on policy is much bigger than in China'.[134] The problem is that there is quite a lot of production but not a lot of feedback.

[125]Wang, *Intentions and Unintended Consequences*, 138.
[126]Ibid., 250.
[127]Interview with bureaucrat B, April 2025.
[128]Zhang et al., "Reforming China's Polar Science and Technology System," 396.
[129]Interview with scholar Q, April 2025.
[130]Interview with bureaucrat B, April 2025.
[131]Wang, "Guojia Zhili Shiye Xia Neican, Neikan de Shiliaoxue Jiazhi."
[132]Xiao, *Beiji Guoji Zuzhi Jian Zhangli Ji Zhongguo Canyu Lujing*, jacket flap.
[133]Interview with scholar F, April 2025.
[134]Ibid.

We also write papers for Beijing, for many different departments in Beijing. But the problem is we don't know who reads our papers at the other end; we don't know who reads them in Beijing, but we actually write quite a lot. Maybe we write two or three papers every month. But we don't know what kind of effect it has on policymaking.[135]

There used to be 'a very small and limited circle of scholars', whereas now, with widespread use of the *neican/pishi* system, 'there is too much input, too many people that are writing memos, reports, so a lot of resources have to be dedicated to screening, to selecting knowledge that might be relevant for policy'.[136]

Producing *neican* is an important part of expert work, with quantitative targets being imposed from above. For example, the SIIS sees 'undertaking national foreign affairs and diplomatic tasks' as a major part of its 'role as a think tank'.[137] According to this report, in 2018, the SIIS contributed with *neican* and research reports on China's Arctic policy to the central government and the Shanghai Municipal Party Committee and Government. In total, the institute provided 212 *neican* in 2018 and 185 in 2017, which was a very positive result for its targets according to key performance indicators.[138] Around 17% of all *neican* produced by SIIS in 2018 received *pishi*, including several from officials from the Treaty and Law Department of the MFA,[139] three from the Standing Committee of the Politburo, and one from Xi Jinping.[140]

Likewise, the Centre for Polar and Ocean Studies at Tongji University reports having more than 30 policy consultation reports they submitted that were adopted or comprehensively adopted by the General Office of the Central Committee of the Communist Party of China, the Shanghai Municipal Education and Health Commission or the Xinhua News Agency.[141] The writing of *neican* is one of the metrics used to evaluate the performance of officials at various levels. In this regard, Zhao demonstrates that local CCP branches need to actively seek out (*zhudong chuji*) scholars willing to provide reports.[142] Provincial research bureaus (*sheng yanjiushi*) are responsible for collecting internal information. The most well-resourced provincial and local authorities may collect reports from researchers in other provinces. For example, the Research Bureau of the People's Government of Shanghai collects information not only from institutions located in the city but also from neighbouring and even more distant provinces.[143]

While communication between experts and the state is mostly unilateral, sometimes the decisions of policymakers come in the form of instructions (*pishi*). *Pishi* is a crucial tool for policymaking that refines it and allows vertically integrated systems to remain flexible. For example, in 1996, China attended the International Arctic Science Committee meeting for the first time, and the Chinese delegation felt that China needed to send an official expedition to the Arctic because 'conducting Arctic scientific

[135]Interview with scholar L, April 2025.
[136]Interview with scholar M, April 2025.
[137]Shanghai Institutes for International Studies, "Shanghai Guoji Wenti Yanjiuyuan 2018 Niandu Xueshu Jiaoliu Gongzuo Jingfei Jixiao Pingjia Baogao," 3.
[138]Ibid., 93.
[139]Ibid., 3.
[140]Shanghai Institutes for International Studies, "Shanghai Guoji Wenti Yanjiuyuan 2018 Niandu Caizheng Zhichu Xiangmu Jixiao Pingjia Jieguo," 6.
[141]Tongji University, "Tongji Daxue Jidi yu Haiyang Guoji Wenti Yanjiu Zhongxin Jianjie."
[142]Zhao, "Dig Deep and Reach Wide."
[143]Interview with bureaucrat B, April 2025.

expeditions is an obligation of member states and a responsibility of a great power'.[144] This status anxiety prompted the SOA to write a report to the State Council recommending that 'at an appropriate time, Arctic research should be elevated to a national undertaking, and national goals for Arctic research should be established'.[145] This report was circulated among CCP leadership, and Zhu Rongji, Li Lanqing, Qian Qichen, and Wen Jiabao all wrote *pishi* on the report.[146] Soon after, the State Council officially approved the establishment of a national scientific expedition team.[147]

Receiving *pishi* is a mark of quality, a source of legitimacy for the chosen institution, and is thus often mentioned prominently on think-tank websites or in scholar CVs. For example, Prof. Li Zhenfu has twice received *pishi* personally from General Secretary Xi Jinping,[148] which means that two of his reports were read by Xi, who either commented or highlighted something in them. Scholars sometimes include in their CVs how many of their policy consultation reports have been 'accepted' as policy (*caina*) or received *pishi*, as well as at which level of the party-state.[149] The Centre for Polar and Ocean Studies at Tongji University boasts that one policy consultation report in 2015 and one in 2016 received positive comments from key central leaders and that one policy consultation report received positive comments from a key leader of the State Council in 2014.[150] Wuhan University boasts that 26 of their polar reports have been accepted for publication by the state and relevant ministries, and eight of them received instructions from central leadership.[151]

Pishi instructions may lead to major policy decisions. For example, in 2016, Xi Jinping issued *pishi* on China's 'polar endeavours', the contents of which are not known but were likely related to the decision to finance construction of Xue Long 2, an icebreaker, as part of the thirteenth Five-Year Plan.[152] Some *pishi* are more symbolic or ritualistic. For example, Li Keqiang, then Vice Premier, issued *pishi* on polar work on important occasions in 2009 and 2011.[153] In 2014, when research icebreaker Xue Long got stuck in ice, Xi Jinping issued 'important instructions' (*zhongyao zhishi*), and Li Keqiang issued *pishi*.[154]

These kinds of *pishi* regarding 'polar work' are widely referenced by the MNR because they are rare events but legitimise work and are able to attract resources. At the same time, *pishi* play an important role in decision-making. For example, in July 1984, a group of 32 polar scientists wrote a letter to the Central Committee of the Communist Party of China and the State Council asking for funding for Antarctic expeditions and construction of China's first Antarctic station. One of the scientists, Xie Zichu, recalled that 'the Central Committee attached great importance to this proposal, and many leaders gave *pishi*', arguing that these *pishi* played a decisive role in establishment of the Great Wall station in 1985.[155] However, as Admiral Liu Huaqing recalls, Great Wall station was

[144]Zhao, "'Huanghe' Yuan Shang Beiji Quan."
[145]Quoted in Yang and Li, "Zhongguo Jinchu Diqiu Liangji de Menhu," 8.
[146]Ministry of Natural Resources, "Tanji Zongheng Ba Wan Li."
[147]Yang and Li, "Zhongguo Jinchu Diqiu Liangji de Menhu."
[148]School of Economics and Management, "Shanghai Haishi Daxue Dishui Hu Jingji Yu Guanli Luntan Xilie Jiangzuo Di 319qi."
[149]Tongji University, "Xia Liping Zhuren Chuxi Quanguo Gaoxiao Guobie."
[150]*Ibid.*
[151]Chutian Metropolis Daily, "Gongguan Chuangxin Cetian Huidi."
[152]State Oceanic Administration, "Haiyangju: Jiakuai Tuidong Woguo Xiang Jidi Qiangguo Maijin."
[153]Li, "Li Keqiang: Puxie Woguo Jidi He Haiyang Kaocha Shiye de Xin Pianzhang"; and *Zhongguo haiyang nianjian 2012*, 5.
[154]Xi Jinping and Li Keqiang, "Xi Jinping Dui 'Xuelong' Hao Shouzu Zuochu Zhongyao Zhishi Li Keqiang Pishi."
[155]Wang, "Xie Zichu: Zhongguo Beiji Kekao Di Yi Ren de Bingxue Rensheng."

already being considered by the State Council and the Central Military Commission (from June 1984) following a request from the SOA, the National Antarctic Research Committee, the State Science and Technology Commission, the People's Liberation Army Navy, and the Ministry of Foreign Affairs.[156] It is thus difficult to definitively conclude that *pishi* directly influence policy outcomes.

Pishi do, however, often influence research funding. In addition to regular research funding, experts might be invited to apply for closed funding calls for commissioned research topics (*keti*).[157] These are often reported as Central Ministerial and Committee–commissioned projects. Unlike general academic grants (like the National Social Science Fund), these are demand-driven and directly serve the immediate policy needs of the government.

The effect could be similar to what happened in Wuhan University when its Antarctic Centre received a congratulatory letter from Xi Jinping. The university then created a permanent exhibition on their polar work, and the centre received additional annual funding of 300,000 RMB, which is significant considering internal competition for funds. The same letter from Xi Jinping features prominently in the university's buildings – it adorns the special exhibition in the university museum and can be seen in the meeting room of the Antarctic Centre, in the museum of the Antarctic Centre, on the big screen in the main hall of the surveying and mapping building, and in the museum of the surveying and mapping faculty. In a similar vein, when Xi Jinping mentioned the construction of the Qinling Station in his 2025 New Year's Address, the PRIC prominently featured the quote on its website.

These practices involve co-producing the policy consultation regime. The party centre signals policy needs vaguely and screens and rewards reports after the fact without specifying its needs. Experts anticipate those needs and propose ways to frame policy problems. The regime rewards accurate anticipation of what the centre will find useful, not specific expert knowledge. The regime is thus a type of boundary infrastructure, one which offers a standardised format through which expertise becomes accessible for the party and through which the party's response is converted back into academic and institutional capital.

Experts between policy and expertise

Chinese experts do boundary-spanning work as they internalise their role as policy advisers. One function that experts have to perform is to 'serve the state' (*wei guojia fuwu*). Numerous scholars make policy proposals and see this as part of their duties.[158] 'If we talk about more core academics, they might serve state interests'.[159] One polar expert, for example, described his role as 'contributing to the state's diplomacy theory and practice' as well as 'providing policy advice and recommendations for the major strategic needs of the nation'.[160]

[156]Liu, *Liu Huaqing Huiyilu*, 503.
[157]Tsai, Li, and Song, "Agent Politics of Chinese Think Tanks."
[158]Interview with scholar J, April 2025.
[159]Interview with scholar K, April 2025.
[160]Zhao, interview by *Pengpai*, August 13, 2025.

Fulfilling this task requires experts to interpret *raison d'état* and translate interests of the state into their work. The main goal of the party-state at present is the second centennial goal, which is defined as the construction of a great power and rejuvenation of the Chinese nation (*qiangguo jianshe, minzu fuxing*). The polar regions are considered part of the 'two priorities and two new initiatives' (*liangzhong, liangxin*) as the development of security capabilities in key areas, including polar regions.[161] Scholar P put it in a very formal manner: 'Xi Jinping has set goals of understanding Antarctica, using Antarctica, protecting Antarctica. And in order to understand the Antarctic, this is precisely where you need scientific research'.[162] The influence of traditional culture should also not be discounted. As one scholar framed it, scholars' relations with the state are part of the Confucian 'three bonds' (*sangang*).[163] They specifically hinted at the reciprocal relationship between ruler and subject, where the former is expected to be benevolent to his subjects and the latter is expected to be loyal. In addition, many scholars are CCP members and perform their duties not just as Chinese nationals but also as party members. This generally creates no dissonance, as the interests of the party and the Chinese nation are considered to be one and the same. Scholar R chooses to compartmentalise their functions: 'The party is about internal politics. I, too, am a party member'.

Chinese polar experts, regardless of their sphere of expertise, are expected to navigate several overlapping fields. This is especially clear for academics, whose primary professional obligation is towards science and academia. For some, the fundamental 'role of scientists is in a way to create a better world'.[164] Generally, 'the role of Chinese scholars is first of all to provide basic knowledge to the whole of society'.[165] The policy advisory role sometimes comes into conflict with these fundamental obligations and snatches time away from more important functions. 'At the universities . . . we only partly work as think tanks, and still have teaching responsibilities and so on'.[166] As scholar R observed, 'Our university doesn't impose on us special [quantitative] requirements [for writing *neican*], and no one has asked me to go write one. . . . My focus is not on writing *neican* but on other research projects'.[167] It is therefore with regret that some scholars remark that it has become more difficult for them to perform these basic functions due to a 'drop in trust' between China and other countries, especially because 'even natural science' is 'being looked at through a security lens', like 'Chinese dual-use science'.[168] Indeed, when asked, most interviewees said they do not consider *neican* as the core of their academic work, even if they are considered a type of academic output. Still, as one scholar put it, along with theoretical articles, media publications, and research reports, *neican* are one of four main types of scholarly output in Chinese academia.[169]

Policy entrepreneurship becomes a necessity because of the opportunities it provides in getting ahead of other institutions.[170] *Neican* count towards publications, which can be used for promotions – for example when applying for professorships. In a bureaucratised

[161]Sanyuan District Development and Reform Bureau, "Shenme shi 'Liangzhong', 'Liangxin?'"
[162]Interview with scholar P, April 2025.
[163]Interview with scholar F, April 2025.
[164]Interview with scholar K, April 2025.
[165]Interview with scholar H, April 2025.
[166]Interview with scholar L, April 2025.
[167]Interview with scholar R, April 2025.
[168]Interview with scholar J, April 2025.
[169]Zhao, interview by *Pengpai*, August 13, 2025.
[170]Ye and Xue, "Motivated to Be Policy Entrepreneurs."

research environment, it becomes one of several instruments in a broader economy of bureaucratic achievement (*zhengji*).[171] Another reason is funding. Competition occurs not only between institutions but within them: units in the same university working within the same discipline compete for the same funding and profile, which incentivises policy entrepreneurship as a means of differentiation.[172] Having a specific expert profile and receiving *pishi* helps attract funding to one's institution. This is important given that Chinese universities face funding cuts to social science and humanities programmes. Scholar M explained:

> To my understanding, the Arctic may have a set of unique research arrangements with traditional research areas, so for example, Tongji University's Centre [for Polar and Ocean Studies] may have a focus on North America, Ocean University in Qingdao may have a focus on Russia, and so on. So … there are already established geographical responsibilities.[173]

In natural science, competition is less about policy influence than about controlling the resources that make research possible. As one researcher remarked, 'In China we say, "data is king" (*shuju wei wang*)', meaning that data is a type of capital for researchers. The same holds for polar expeditions. Since expedition participation is capped at one or two members per institution, selection reproduces existing imbalances. Places are distributed according not only to research requirements but also to the institution's administrative subordination in the maritime *xitong*.[174] It is also gendered, and women remain a minority in Chinese National Antarctic Research Expedition rosters. This is partly due to informal bias, where Chinese women scientists describe being considered too 'weak to take on harsh weather conditions and tragedies' of Antarctic expeditions.[175] Official media frames women's expedition participation in gendered-complementary terms.[176] The gendering of polar careers also extends into researchers' lives. For example, women can apply for youth grants in social and natural sciences until age 40, five years longer than men, justified by 'the burden of child care' and the party-state's pronatalist policy.[177] These accommodations operate under and reinforce the assumption that women bear family responsibilities. Still, similar adjustments have not been made to tenure timelines.[178] Career advancement thus comes into direct conflict with personal life.[179]

In recent years, there has been a growing trend towards establishing new regional studies centres at various universities, driven by funding and China's growing need for foreign policy expertise. Sometimes these centres may even acquire the required expert profile after the fact. For example, in 2017, Soochow University founded the Centre for Lao Studies. Scholar M interpreted this decision specifically as the university's desire to better position itself in competition with other institutions. Wang has documented that one privately funded, undergraduate-only college joined the China–Nordic Arctic Research Centre with the explicit aim of raising its domestic visibility, improving its

[171]Wang, *Intentions and Unintended Consequences*, 214.
[172]Interview with scholar O, April 2025.
[173]Interview with scholar M, April 2025.
[174]Zhang et al., "Reforming China's Polar Science and Technology System," 397.
[175]Guo and Gupta, "Chinese Women Set to Conquer Antarctica, Gender Inequality."
[176]Wang, "Female Power Pushes the Limits."
[177]*China Daily*, "Age Limit Raised to Give Women Better Support."
[178]Gao et al., "Academic Mothers in Chinese Universities."
[179]Bao and Wang, "I Am Willing to Do Both Well."

institutional reputation, and building ties with more prestigious institutions.[180] Its admission generated discontent among other Chinese members and contributed to tighter membership criteria the following year.[181]

Chinese polar experts are boundary spanners, as they do not hold a stable position on one side or the other of a line between science and policy. They have several roles at once: researcher, adviser, party member, and competitor for berths and for data. The boundary is not a line they cross but a set of obligations they carry simultaneously, and the *neican/pishi* system is one such boundary infrastructure within which they meet those obligations.

Conclusion

Polar regions present a highly technical field, where participation in governance requires expertise in fields from natural science to international law. As other articles in this special issue show, experts with diverse specialisations are often directly involved at various stages of policymaking. China is not unique in this regard. For example, foreign policy think tanks in the U.S. have a significant impact on legitimising policy, agenda-setting, and policy formulation.[182] What is unique to China are the specific mechanisms of expert involvement, not only through policy advisory conduits but also through party channels.

This article has sketched out three practices in which polar expertise enters policy-making in China. These correspond to different kinds of boundary work. Horizontal advisory committees are boundary organisations mandated to coordinate expertise across ministries, and their public profile has faded since the mid-2010s. Documents like the Arctic Policy White Paper are boundary objects: the centre mobilises expertise ad hoc to fill gaps in in-house expertise, allowing concepts like 'near-Arctic state' to enter official policy. The White Paper was designed as and serves as orthodoxy for China's official position on the Arctic. The unique Chinese internal reporting is a boundary infrastructure that converts expertise into legible, routine policy feedback. Academics who write policy reports are boundary spanners: they are researchers who must maintain their academic credentials, but they also sometimes act as bureaucrats or policy advisers who contribute to the policymaking process and even participate in Track II diplomacy.

The polar case illustrates co-production under mutual uncertainty. The party-state centre is unclear about what knowledge it needs on polar regions, and experts anticipate those needs but are not always certain how this advice is used. Ye and Xue show that tighter control of public debate and the drive to build think tanks have made report-writing a rewarded activity and turned many Chinese academics into policy entrepreneurs, but Ye and Xue treat experts' guessing about knowledge needs as an inefficiency.[183] The polar case shows that the centre is not always capable of specifying its needs in advance, so the guessing is a feature of the system which rewards accurately predicting what the centre will find useful.

[180]Wang, *Intentions and Unintended Consequences*, 214.
[181]Ibid., 253.
[182]Zaytsev, Kuskova, and Kononova, "The Power of Knowledge."
[183]Ye and Xue, "Motivated to Be Policy Entrepreneurs," 7, 9, 11.

At the top, polar affairs fall to the Central Foreign Affairs Commission, where they compete with every other foreign policy issue for attention. The centre does not orchestrate polar knowledge production from a fixed view of its interests. It signals broadly and selects advice from experts competing to supply some of the framing of polar issues. This means that scholarly discourse is a menu of possible framings, some of which, like 'near-Arctic state', are later adopted. These findings qualify the image of China's polar policy as coherent and centrally orchestrated. Authority over polar affairs is fragmented within and between ministries.

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Appendix

Type of interviewee	Code	Information about interviewee
Bureaucrat	A	Manager at a research institute
	B	Official at a local party-government branch
Scholar	C	Research institute, social science
	D	University, social science
	E	Research institute, social science
	F	University, social science
	G	University, social science
	H	Research institute, social science
	I	Research institute, social science
	J	Research institute, social science
	K	Research institute, social science
	L	University, social science
	M	Research institute, social science
	N	Research institute, social science
	O	University, natural science
	P	University, natural science
	Q	University, social science
	R	University, social science
Consultant	S	Private consultancy